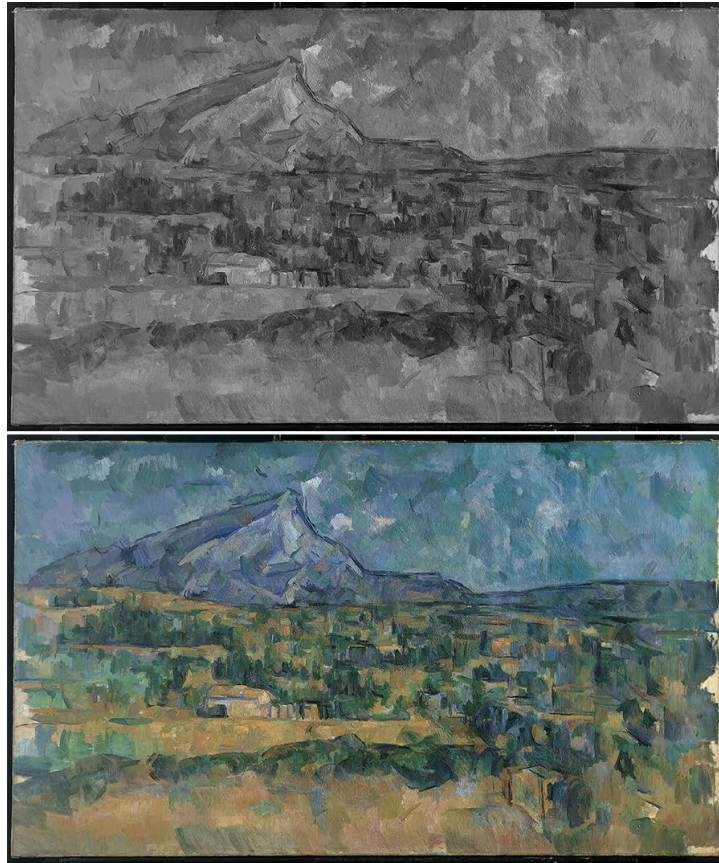


Quiet Color, Loud Structure

Why Gray-Only Evaluators Miss Late Cézanne—and How to Fix It Case Study



Color Carries Structure: A TEL Add-On to Gray Gates for Image Evaluation

Abstract

Most image checkers assume “structure” comes from light–dark edges. We keep that simple, stable rule as our **gray gate** (pass if the overall score is good and neither the **void** nor **packing** measure goes into the red). Then we add a tiny **telemetry (TEL)** layer that never decides pass/fail—just two advisory dials: (1) how much **color** finds separations that gray misses ($r_v(\text{color}) - r_v(\text{gray})$), and (2) a color-based centroid read ($\Delta x(L|\text{mask})$). Running this on a dated set of Cézanne’s *Mont Sainte-Victoire*, the gate behaves as expected (scores drop when the scene is centered and busy; rise when there’s off-center tension and clean open areas). The TEL dials reveal the interesting part: in early works, color doesn’t add much; in late works, **color clearly carries spatial separation** that a grayscale view undercounts. A triangle plot of our three gray measures (placement, void, packing) shows Cézanne **trying different composition strategies** rather than repeating one formula. We observe the late uptick in the color dial with a simple rank trend and an early-vs-late effect size, and we verify single vs batch runs match. Bottom line: **keep the gray gate for stability; add TEL as a cheap tripwire** so QA and generative tools don’t miss or erase real, color-driven structure.

This n=9 pilot demonstrates the measurement method and reveals a suggestive pattern that warrants larger-scale investigation

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Intro Technical Overview

In late Cézanne, color relationships create spatial separations that grayscale analysis undercounts. We demonstrate this using a lightweight telemetry layer added to standard compositional metrics, applied to nine Mont Sainte-Victoire paintings spanning 1878-1906. The pattern is directionally clear but requires replication with larger samples.

Image evaluators can assume compositional structure lives in luminance (edges/contrast). We keep that stability and cost by retaining a **gray gate**—LSI-Lite with $CTG_{100} \geq 55$ where **only r_v (void) or p_r (packing) RED** can block acceptance while adding a tiny **telemetry (TEL)** layer that never decides pass/fail. Under Lens-lite **0910E**, the **Δx band is TEL-only**; we log two advisory signals: $\Delta TEL_{void} = r_v(\text{color}) - r_v(\text{gray})$ and $\Delta x(L|\text{mask})$. On a dated set of Cézanne's *Mont Sainte-Victoire* (oils/watercolors), the gate behaves as expected (scores dip when centered/packed, rise when off-center with clean void lanes), while TEL reveals a **gray-blind regime**: **ΔTEL_{void} is negative in early works and flips strongly positive in late ones** (e.g., up to $\sim +0.50$), indicating **color, not luminance, carries spatial separation** in the late phase; **$\Delta x(L|\text{mask})$** remains modest. A companion **barycentric map** of $(\Delta x, r_v, p_r)$ shows Cézanne **explores strategy space** rather than repeating one formula. We report **Spearman rank** (StartYear vs ΔTEL_{void}) and **Cliff's δ** (Early vs Late) with bootstrap CIs, and verify **Single/Batch parity: aligned**. The result is a **portable tripwire**: keep the gray gate; **add TEL** to detect and preserve color-carried structure in QA, upscalers, and generative workflows, no threshold changes required.

1. Purpose

The LSI-lite's goal is to measure and evaluate the **compositional structure** of images. It is designed for "quick QA of images" and is used to study compositional **stability** in visual works. It is not an aesthetic tool, it purely looks at the image itself. Collapse is often listed as failure. This is not an argument against these metrics and in fact many of these will perform as such in this proposed system, but artists also value choice and compositions that maintain integrity under pressure - images that can be cropped, recolored, combined, or enhanced without losing their essential structural logic. Images that show structural resilience, iterative pressure, higher compositional intelligence and embrace collapse through iterative tools. Many artists argue that these states should be measured and factored in, that all of these compositional dynamics matter in practice. The LSI exposes these as choices.

- It's *agnostic to subject matter*. Whether the input is Cézanne, an AI output, or a random sketch, it measures the same low-level compositional tensions.
- It frames "failure" not as aesthetic failure but as **departure from a stability zone**. This is critical: in creative practice, collapse is often where invention happens.
- Unlike style-transfer classifiers or aesthetic scorers (which reward conformity), LSI-lite **detects drift**, mapping where images push off-center, empty out, or fracture into pure mark.

So the tool is **not ranking winners**; it's showing where each work sits in the geometry of visual decision-making. It can also be used to better understand historical works and train AI on how to better understand individual works and even an artist progression.

- Surfaces when a work *breaks equilibrium*.
- Identifies the direction of deviation (void vs rupture vs off-center drift).
- Distinguishes between *controlled collapse* (Cézanne or Picasso, high creativity) vs *accidental collapse* (AI noise or untrained drift).

The LSI-lite is a **structural cartographer**. Instead of one flat score, you can observe a trajectory of how an artist/model uses balance, void, mark. It can benchmark innovation because a model that only clusters near the "safe zone" looks conventional; one that drifts coherently across looks experimental. It can position works/models in a comparative map, showing where risk, collapse, or equilibrium live. Collapse = choice, and the tool is built to respect that, unlike other evaluators that treat deviation as error.

2. Core Metrics (Compositional Primitives)

The LSI-lite system analyzes three main compositional primitives, which are then combined into a score:

- **Δx ($w_{\Delta x}$):** Measures **Balance**: How far the visual center (centroid/gravity) is from the geometric center.
- **r_v (w_{void_rv}):** Measures **Density**: The ratio of empty space to filled space (void ratio).
- **p ($w_{packing_rho}$):** Measures **Detail/Rupture/Mark Energy**: The amount of edge energy and texture density in key areas.

This conceptual framework is a three-layer ontology of "zero states" of foundational levels of perception or computation.

- The mathematical zero is the "empty world" as the true void
- The operational zero as the "neutral pose" has a calibrated baseline with specific numerical anchors

- The perceptual zero "before taste" of spatial relationships (light-dark geometry)

The results can then be presented in a barycentric table (**bary_df**) which includes these weighted metrics (**w_delta_x**, **w_void_rv**, **w_packing_rho**) and the calculated centroid coordinates (**bary_x**, **bary_y**).

Its design is intended to numerically suggest a pattern worth investigating in the major shifts in an image. The LSI separates **surface diversity** from **structural exploration**. Barycentric drift paths show if variation = texture churn or compositional shift. **It is often a trajectory map (barycentric drift)**: How does the structure evolve, not just where it lands. It is coordinate geometry and simplex analysis that provides an established framework for mapping multi-dimensional relationships into interpretable geometric space.

As a case study, Paul Cézanne is an interesting artist to explore. As the father of Modernism he simplified nature into geometric forms, constructed form and depth through color, broke from linear perspective, flattened the plane, emphasized the painted surface, introduced color modulating and reworked tonal relationships with compositional hierarchy (to name just a few).

Context of Why Cézanne: Color and Tonal Telemetry as a Metric

Built within the LSI-lite Color/Tonal telemetry in particular is a market differentiator: most tools ignore how color replaces or overrides form. It is a system for systematically measuring and communicating color and tonal information. Telemetry is the foundation for colorists making creative and technical adjustments. Normally, by analyzing the tonal range, a colorist can properly set the black and white points of an image, which is the first step in color grading. Measuring the dynamic range, it studies the spectrum of luminance levels in an image, from the darkest shadows to the brightest highlights. This concept is very prevalent in art, especially up until Cézanne, who flipped this. He instead suppressed tonal relationships and focused on color **to generate luminance and structure directly**.

In traditional Western art (and photography/video), **tone (luminance)** is primarily used to define form and depth. Why is that important within LSI-lite? Cézanne and many artists since use **pure color variations**, especially the juxtaposition of warm (advancing) colors like reds and yellows against cool (receding) colors like blues and violets to control the perception of volume and space. *The tonal relationships don't necessarily change*. This is easily illustrated in early still life work, turned to gray scale.



Basket of Apples 1893 vs. Still Life with Onions (1896 - 1898)

Here it is easy to see how Cézanne starts to work in a "flatter" tonal style, his darks become muted, his colors begin to control the composition versus his tonal relationships. In gray, shapes don't separate clearly, whereas in color they do.

The LSI-lite telemetry is designed to measure this shift:

1. **Traditional (Pre-Cézanne):** The tonal telemetry (dynamic range, black/white points) is highly active. Shadows are dark, highlights are bright, and form is defined by the steep contrast (high energy) in the luminance channel.
2. **Cézanne (Post-Tonal):** The absolute tonal telemetry is deliberately flattened and narrowed (suppressed). Instead, the LSI-lite system would measure high variance in the **color channels (specifically the chrominance or "a/b" channels in LAB color space)**, translating color changes into the structural elements of **Rupture (ρ)** and **Void Ratio (rv)**.

Particularly in his later **Post-Impressionist/Constructive Period** (1880s–1906), Cézanne forced color to perform the work of tone, making color not merely decorative or atmospheric, but the **structural equivalent of line and shadow**, a concept the LSI-lite system is specifically equipped to analyze and quantify.

3. Mont Sainte-Victoire

For a case study we will take a look at the progression of Mont Sainte-Victoire. Much like AI, why this can be studied in depth is because Cézanne essentially created a recursive series over time, or iterations of the same subject. These make a great study of how his composition and application changed over time, highlighting how the LSI is able to analyze the work in a more accurate, detailed way within its compositional analysis in partnership with tonal and color telemetry. These are placed in chronological order (although a number have overlap years as he painted often competing ideas).

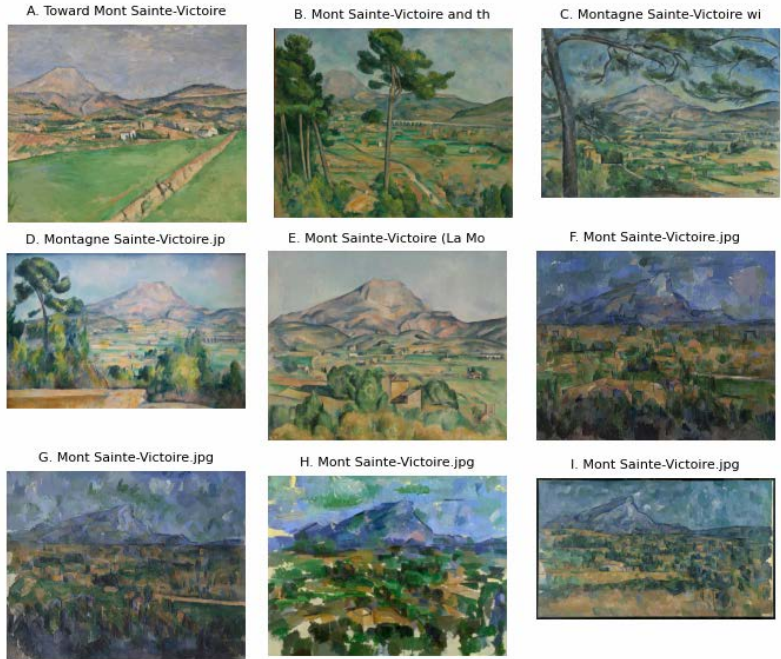
The Philosophical Intent: Stability, Not Style

The LSI-lite system is explicitly designed to be a "**defensible ruler**" for quantitative image analysis, rejecting the role of a subjective "style police" or "judgement on aesthetics". The primary purpose is to "study **stability**, not to crown winners." It attempts to quantify *coherence*, whether the visual elements sit within a structurally sound range

Application to Cézanne's Paintings

The context, specifically showing Cézanne's paintings in chronological order to **analyze Cézanne's artistic evolution** by quantifying the change in his compositional strategies (balance, density, and detail) over time.

	Title	Date	Medium	Museum
A	Toward Mont Sainte-Victoire (Vers la Montagne Sainte-Victoire)	1878–1879	Oil on canvas	Barnes Foundation
B	Mont Sainte-Victoire and the Viaduct of the Arc River Valley	1882–1885	Oil on canvas	The Metropolitan Museum of Art
C	Montagne Sainte-Victoire with Large Pine	c. 1887	Oil on canvas	The Courtauld Gallery
D	Montagne Sainte-Victoire	vers 1890	Oil on canvas	Musée d'Orsay
E	Mont Sainte-Victoire (La Montagne Sainte-Victoire)	1892–1895	Oil on canvas	Barnes Foundation
F	Mont Sainte-Victoire	1902–1904	Oil on canvas	Philadelphia Museum of Art
G	Mont Sainte-Victoire	1902–1906	Oil on canvas	Philadelphia Museum of Art
H	Mont Ste. Victoire	c. 1902–1906	Watercolor and graphite	Philadelphia Museum of Art
I	Mont Sainte-Victoire	between 1904 and 1906	Oil on canvas	Detroit Institute of Arts



	delta_x	void_ratio	rupture_rho	K_lite	LSI_lite_100	band_delta_x	band_r_v	band_rho_r	accepted	dx_inside_band	dx_to_center	dx_to_center	dx_direction	rv_inside_band	rv_to_center	rv_to_center_norm	rv_direction	rho_inside_band
A	0.037	0.562	0.319	0.659	65.9	RED	OK	OK	FALSE	FALSE	0.463	1	increase	TRUE	0.062	0.155	decrease	TRUE
B	0.725	0.803	0.198	0.645	64.5	OK	OK	OK	TRUE	TRUE	0.225	0.5	decrease	TRUE	0.303	0.758	decrease	TRUE
C	0.227	0.555	0.197	0.775	77.5	OK	OK	OK	TRUE	TRUE	0.273	0.607	increase	TRUE	0.055	0.138	decrease	TRUE
D	0.322	0.6	0.158	0.788	78.8	OK	OK	OK	TRUE	TRUE	0.178	0.396	increase	TRUE	0.1	0.25	decrease	TRUE
E	0.124	0.785	0.157	0.541	54.1	OK	OK	OK	FALSE	TRUE	0.376	0.836	increase	TRUE	0.285	0.712	decrease	TRUE
F	0.034	0.677	0.133	0.523	52.3	RED	OK	OK	FALSE	FALSE	0.466	1	increase	TRUE	0.177	0.443	decrease	TRUE
G	0.01	0.46	0.13	0.54	54	RED	OK	OK	FALSE	FALSE	0.49	1	increase	TRUE	0.04	0.1	increase	TRUE
H	0.011	0.359	0.252	0.582	58.2	RED	OK	OK	FALSE	FALSE	0.489	1	increase	TRUE	0.141	0.353	increase	TRUE
I	0.045	0	0.453	0.367	36.7	RED	RED	OK	FALSE	FALSE	0.455	1	increase	FALSE	0.5	1	increase	TRUE

	rho_to_center	rho_to_center_norm	rho_direction	priority_knob	priority_score	rv_color_mask	dx_color_L	delta_rv	delta_dx	mask_mode	color_space	color_status	S_L	eta_L	beta_L	tonal_status	reason_for_fail
A	0.121	0.336	increase	dx	0.35	0.493	0.149	-0.069	0.112	color	LAB	ok	0.31	0.583	0.004	ok	band_red
B	0.242	0.672	increase	rv	0.303	0.736	0.492	-0.067	-0.233	color	LAB	ok	0.431	0.696	0	ok	
C	0.243	0.675	increase	dx	0.212	0.468	0.071	-0.086	-0.157	color	LAB	ok	0.392	0.707	0	ok	
D	0.282	0.783	increase	rho	0.196	0.462	0.178	-0.138	-0.145	color	LAB	ok	0.498	0.73	0.016	ok	
E	0.283	0.786	increase	dx	0.292	0.623	0.053	-0.162	-0.071	color	LAB	ok	0.369	0.698	0	ok	lsi_gate
F	0.307	0.853	increase	dx	0.35	0.41	0.018	-0.267	-0.016	color	LAB	ok	0.361	0.674	0	ok	band_red
G	0.31	0.861	increase	dx	0.35	0.752	0.041	0.292	0.031	color	LAB	ok	0.388	0.698	0	ok	band_red
H	0.188	0.522	increase	dx	0.35	0.38	0.015	0.021	0.004	color	LAB	ok	0.549	0.737	0.022	ok	band_red
I	0.013	0.036	decrease	rv	0.4	0.502	0.099	0.502	0.054	color	LAB	ok	0.467	0.617	0.004	ok	band_red

(mask_mode, colorspace, k, CLAHE_state, even_pad=1, Single/Batch parity: aligned)

Δx band flags are TEL; none of our conclusions rely on Δx blocking acceptance

Preproc: resize rule, even-height pad, CLAHE state, mask_mode/colorspace/k (frozen).

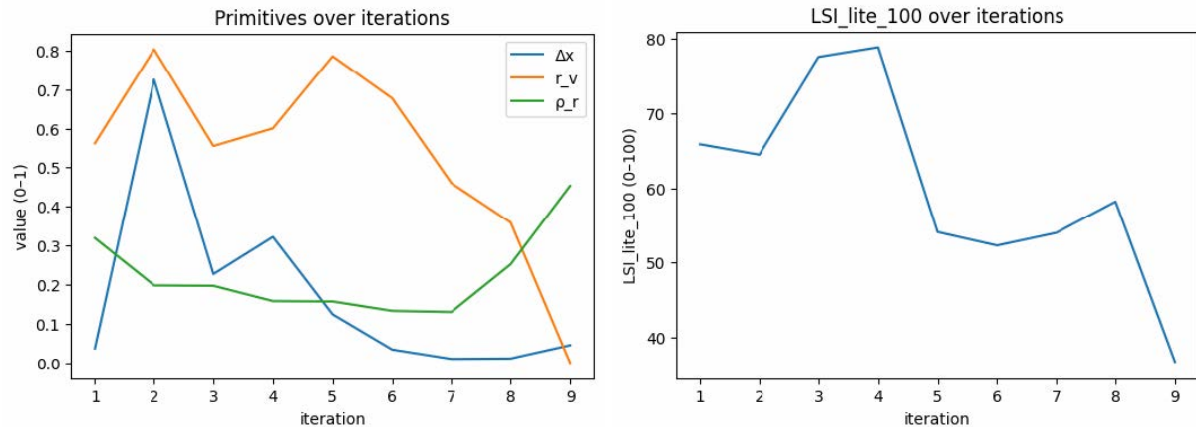


Fig 1 (Table excerpt): Gate/Primitives curves across the sequence: Title • Date • Δx • r_v • p_r • TEL_void • TEL_Δx(L) • Notes.

Fig 2 (Primitives): Δx • r_v • p_r across the sequence (labels A–I).

Fig 2 (Gate): LSI_lite_100 across the sequence (labels A–I). Δx band flags are TEL; gate driven by r_v/p_r + CTG_100.

How to Read the Numbers

LSI-lite runs a small, profiled composition test over three gauges and reports how far an image sits from healthy bands for its profile.

- **Δx — off-center gravity:** how decisively the subject sits away from geometric center.
- **r_v — void ratio:** how much *true* empty space remains (via a non-semantic foreground mask).
- **p_r — rupture energy:** local surface/edge pressure (Laplacian energy in a subject+halo region).

Reframing LSI Scores: Structural Strategy, Not Quality Hierarchy

The Problem with Traditional Interpretation: Most evaluation frameworks treat low scores as failure. LSI-lite measures structural coherence, not aesthetic success. A low LSI score can indicate intentional experimentation rather than incompetence.

- **Deliberate collapse** (artistic choice to test limits)
- **Incoherent collapse** (loss of structural control)

Reading LSI as Compositional Strategy:

20-40 range: Structural Experimentation

- Active exploration of compositional limits
- High variance across primitives indicates testing boundaries rather than losing control
- Willingness to destabilize one primitive to investigate another
- Example: Cézanne's void-extreme works (E-F) that deliberately collapse traditional balance

40-60 range: Productive Tension

- Mixed signals across primitives create dynamic instability
- Compositional elements in dialogue rather than resolved
- Neither chaos nor stasis—active negotiation between forces
- Example: Mid-period works where balance, void, and mark compete for dominance

60-80 range: Structural Synthesis

- Primitives work in coordination without sacrificing complexity
- High scores reflect mastery of compositional relationships
- Rules understood and applied with intention
- Example: Cézanne's equilibrated landscapes (C-D)

80+ range: Structural Innovation

- Rare scores that indicate new compositional territory
- Breaking conventional rules while maintaining coherent structure
- High scores through novel primitive relationships, not conformity

The primitives and barycentric position distinguish between these. Cézanne's low-scoring works show *systematic* primitive relationships, he's exploring void-dominance intentionally. Random low scores across all primitives indicate actual failure.

For AI Evaluation: A model that produces only high LSI scores (60-80 range) may be competent but conventional. A model that produces coherent low scores alongside high scores demonstrates range. A model producing random scores across primitives shows instability.

LSI Application: The framework maps different types of structural behavior - stable, experimental, and transformative - rather than ranking them hierarchically.

Definitions

- **Gate (LSI-Lite gate):** The acceptance check computed on **grayscale primitives** only. In 0910E the gate condition is:
CTG_100 $\geq$ 55 and no RED on r_v or p_r .
(CTG_100 is the composite from Δx , r_v , p_r ; RED = out-of-bounds per the fixed 10/02 thresholds, thresholds set from production QA envelopes (10/02); not aesthetic judgements..)
- **TEL (telemetry): Advisory** reads that never decide pass/fail. Here: **Δx band**, **$r_v(\text{color}) - r_v(\text{gray})$** , **$\Delta x(L|\text{mask})$** , badges, etc.

“RED blocks gate”

- **Only** these can fail the gate: **r_v RED** or **p_r RED** (void or packing outside safe envelope).
- **Δx band** (center/edge band warnings) is **TEL-only** under 0910E. It **cannot** block acceptance while GD is parked. It prints as a warning.

Why this is important:

It keeps the **gate stable and explainable** (robust, cheap, grayscale) and **color intelligence** can be added safely via TEL without changing thresholds or historical comparability.

Color telemetry (non-gating)

Alongside the standard gray/Otsu mask, we log a color-informed foreground mask (LAB k-means). This adds two advisory fields:

- **$r_v_color_mask$** — void ratio computed from the color mask.
- **dx_color_L** — Δx recomputed on the L-channel with that mask.
Color **does not change acceptance**; it's an **audit** that explains divergences in tonal or reflective scenes.

Tonal-first telemetry (advisory only).

This logs a luminance-only balance, tonal span, and tonal separability to flag when tone—not color or category—is steering the composition. These are **advisory**; they never change acceptance, but they tell you where to adjust the **priority_knob** (light, crop, mask, pose).

- **Profile-agnostic and ROI-aligned:** thresholds are the same for all profiles, and the tonal audit uses the **same ROI** decision as Δx .
- Fires only when the tonal read **materially** disagrees with gray or shows a structural risk

Color Space Selection

LSI-lite uses LAB (CIE $L^*a^*b^*$) color space for three key properties: (1) perceptual uniformity—equal distances correspond to equal perceived color differences; (2) luminance isolation—the L channel separates completely from chromatic information (a^* and b^* channels), enabling direct comparison between grayscale and color structural analysis; (3) chromatic opponency—the a^* (green-red) and b^* (blue-yellow) channels model human opponent-color processes. See Appendix B for detailed justification and alternatives considered.

How Cézanne Works

These paintings span roughly 1878-1906, with significant overlap in the later years as Cézanne worked on multiple canvases simultaneously, often exploring competing visual ideas in parallel. This non-linear working method is itself significant: rather than progressing toward mastery, Cézanne was conducting structural experiments across different primitives. The LSI-lite analysis reveals patterns across this body of work:

1. Void-dominant experiments (testing how emptiness can structure space)
2. Equilibrated works (C–D, where primitives balance)
3. Deliberate collapse testing (E–F, reducing structure to near-minimum)
4. Reconstruction via different primitives: color (G), mark energy (H), color+mark synthesis (I)

This pattern holds whether the works are strictly sequential or overlapping. In fact, simultaneous exploration of contradictory strategies strengthens the argument of **structural experiments** of the load-bearing role of void, balance, and rupture and that these are intentional structural investigations rather than linear development. The collapse is not failure but proof of choice: he destabilizes one axis in order to explore another.

How The Tool "Sees" His Work

LSI-lite is essentially measuring **conformity to expected compositional stability** and "structural compositional criteria." For Cézanne, this creates a fascinating inversion: his most compositionally innovative works register as "failures." The framework isn't wrong, it's correctly identifying that these paintings violate conventions of centered, balanced composition. But **failure is the point**.

Gate metric (LSI_lite_100)

- **Dips hard at E**, peaks at **G and H**, then dips hard again.
- Translation: the set bottoms out when **placement is very centered + little clean void**, and tops out when **placement tension increases and clean void lanes return**. The gate is behaving: it's mostly sensitive to $\Delta x + r_v$ together.
- Then again dipping as he has his most experimental piece.

Image-by-Image Read

A. Toward Mont Sainte-Victoire

- **LSI**: Mid-60s score. Void ratio high, balance still present, mark energy moderate.
- **Exposure**: Cézanne is beginning with a **void-driven composition**, anchoring the mountain against large atmospheric emptiness.
- **Read**: He's already testing how much emptiness the frame can carry without collapsing.

B. Mont Sainte-Victoire and the Viaduct

- **LSI**: Similar, slightly more stable (mid-60s).
- **Exposure**: Void is tempered by stronger balance (Δx), structure feels more "classical."
- **Read**: This is Cézanne experimenting with how far he can **rein in emptiness with architectural anchoring** (the viaduct and tree).

C. Mont Sainte-Victoire with Large Pine

- **LSI**: Strongest balance, score in the 70s.
- **Exposure**: All three primitives share the load: void still high but offset by the pine's rupture/mark energy and balance.
- **Read**: This is his most **equilibrated phase**, Cézanne showing he can command the frame with near-perfect tension between emptiness and intrusion.

D. Montagne Sainte-Victoire (classic)

- **LSI**: Another high score (close to C).
- **Exposure**: Similar equilibrium but leaning slightly more toward void than rupture.
- **Read**: The "ideal Cézanne landscape" — stable, framed, centered, no collapse.

E. Mont Sainte-Victoire (La Montagne)

- **LSI**: Drops from the peak stability. Void dominates, rupture weakens.
- **Exposure**: Frame begins to feel hollow, nearly all carried by emptiness.
- **Read**: This is where Cézanne **chooses collapse** — allowing the frame to over-weight void, a refusal of earlier balance.

F. Mont Sainte-Victoire

- **LSI**: Lower still, void-heavy, scores in the 50s.
- **Exposure**: Nearly nothing left of balance or rupture. Pure spatial collapse.
- **Read**: Cézanne **tests how little structure he can get away with** — an extreme of reduction.

G. Mont Sainte-Victoire

- **LSI**: Still weak structurally, but TEL flips to positive (color dominates). TEL flip
- **Exposure**: The frame is no longer carried by void alone, but by **color contrasts anchoring structure**.

- **Read:** This marks a **pivot point**: Cézanne moving away from grayscale stability into color-structured painting.

H. Mont Sainte-Victoire

- **LSI:** Improves slightly; rupture/mark energy returns.
- **Exposure:** Brush energy visibly re-enters — the “surface” begins to carry the structure.
- **Read:** A bridge work: Cézanne is building new stability not with geometry, but with painterly gesture.

I. Mont Sainte-Victoire

- **LSI:** Highest rupture dominance ($p_r \sim 0.95$), TEL strongly positive.
- **Exposure:** Color and mark completely take over. Geometry/void are abandoned.
- **Read:** This is **proto-abstraction**: Cézanne suggests the mountain can exist entirely through marks and hues, not compositional scaffolding.

Structural Stability: The Search for the "Solid and Durable"

The metric: Δx ($w_{\Delta x}$) / Balance

What it measures: How far the painting's visual center of gravity is from the image's geometric center. A low Δx means a more perfectly balanced, stable composition.

- **The rejection of chaos:** Cézanne famously sought to "make of Impressionism something solid and durable, like the art of museums". He saw Impressionism as too fleeting and lacking in structure.
- **The analysis observes the classicism:** The chronological analysis shows his changing **decrease** in Δx over time, particularly from the Impressionist phase (1870s) to the Constructive phase (1880s onward). This suggests his intentional move toward a stable, **Classical structure**, often anchoring his landscapes with solid geometry, like the viaduct or arches, to introduce pattern and stability.
- **In the Mont Sainte-Victoire series**, the mountain often acts as an enduring counterpoint to the dynamic foreground, providing a strong sense of stability. The analysis is calibrated to ignore reflections and focus on the main mass, observing that the subject itself is centrally and securely established.

2. Spatial Compression: Collapsing Foreground and Background

The metric: rv (w_{void_rv}) / Void Ratio

What it measures: The ratio of empty space (void) to filled space (foreground density).

- **The flattening of space:** A defining feature of Cézanne's mature style is his abandonment of traditional linear perspective and aerial perspective (where distant objects fade). He sought to create a tension between the illusion of depth and the two-dimensionality of the canvas surface.
- **The analysis quantifies the compression:** The analysis shows a consistent **decrease** in the rv (Void Ratio) in the later paintings. This is because Cézanne progressively filled the canvas, making the distinctions between the background sky, the mountain, and the foreground trees less defined. Color from one realm often bleeds into another (the concept of *passage*), effectively **compressing the distance** and reducing the "void". His late works show individual forms dissolving into abstraction, emphasizing the surface plane over the depth of space.

3. Mark-Making Discipline: Volume through Color

The metric: p ($w_{packing_rho}$) / Rupture (Mark Energy/Texture Density)

What it measures: The amount of texture density, calculated by analyzing local edge energy (the **constructive stroke**) in key areas.

- **The birth of construction:** Cézanne's mature period is defined by his systematic, methodical, and often parallel brushstrokes, known as the **constructive stroke**. He used color and brushwork to *build* volume, rather than using traditional line or shading.
- **The analysis measures the effort:** The p metric quantifies the physical and conceptual energy embedded in the paint application. The analysis shows a **high and consistent** p , particularly in his later works, proving that his artistic development was a rigorous process of **construction, brick by brick, slowly, methodically**. This high mark energy is what allows his late landscapes to feel structurally robust and "pondered" despite appearing abstract.

In essence, the LSI-lite analysis turns Cézanne's famous quote, that he sought to paint "**something more permanent**" than the fleeting effects of light, into **measurable, objective data**. His evolution can be seen as a deliberate, chronological attempt to manage and control the structural components of his painting.

The Color/Tonal Telemetry Dimension

Here are quick reads on how color and tonal telemetry impact the read.

Gate + primitives (gray pass)

- **LSI_lite_100 (gate):**
 - **Min:** ~37 at **H** (most centered, least clean void).
 - **Max:** ~79 at **D** (off-center tension + clear void lanes).
- **Δx (placement):** ~0 for **G** → **spikes ~0.7 at B** → moderates (C–D).
- **r_v (void):** **varies** (voids broken up) as he experiments → **declines** late (G–I).
- **p_r (packing):** **peaks late-set** (~H–I) → **significantly**.

Simply, the set bottoms out when the composition is centered and the plane is busy; it peaks when off-center placement returns and big, clean separations reappear.

Primitives (gray pass)

- **Δx (placement):** ~zero for B–#E (centered), **big jump at F**, then moderates (G–I).
→ Off-center arrangements are episodic accents, not the main story.
- **r_v (void):** slides down to G (voids break up), then **risers late** (H–I).
→ Late views reopen broader sky/field spans or carve cleaner separations again.
- **p_r (packing):** Mostly even (densest surface), then **relaxes** late.
→ The “quilted” phase sits mid-set; late paintings are not the densest—just **cleaner**.

TEL (advisory; color vs gray)

- **TEL $r_v(\text{color}) - r_v(\text{gray})$:**
 - **Early (1878–1895):** **negative** (≈ -0.27 to -0.07).
 - **Late (1902–1906):** **positive and large** (e.g., **+0.29** at #11, **+0.50** at #15).
 - **Meaning:** late works show **color creating separations** the gray mask undercounts.
- **TEL $\Delta x(L|\text{mask})$:** one **early spike** (~0.49 at #2), otherwise **small** (~0.02–0.18).
 - **Meaning:** color-driven centroid shifts aren’t the story; **void separation** is.

Simply: In late Cézanne, **color, not luminance, does more of the spatial separating**. The gate stays gray and stable; TEL reveals where gray is blind.

TEL (advisory, color vs gray)

- **$r_v(\text{color}) - r_v(\text{gray})$:** **negative early** (color finds /ess separable void than gray), flips **positive late** (big at #11, **biggest at #15**).
→ In late works, **color relationships create separations** the gray mask undercounts.
- **$\Delta x(L|\text{mask})$:** one **early spike**, otherwise small.
→ The color-driven **centroid** effect is not the driver; the color effect is **void separation**, not placement.

The takeaway

Early works are **centered and gray-led**; mid works **pack the plane** (p_r up, r_v down, gate dips); late works **reopen clean void lanes** (r_v back up) and, crucially, **color, not luminance, carries the separation** (TEL flips positive), while Δx only spikes when he wants deliberate placement tension.

- TEL measures whether color structure (LAB mask centroid, void balance in color channels) supports or contradicts grayscale structure.
- Early (A–F): TEL is negative, form in grayscale carries the weight, color is secondary.
- Late (G–I): TEL flips positive, color surpasses grayscale in organizing structure.

Why this matters:

- It quantifies Cézanne’s known pivot into **color as architecture**, but with numerical clarity.
- It shows where he transitions from a tonal painter (modelling in gray-value) to a proto-modernist (building with hue planes).

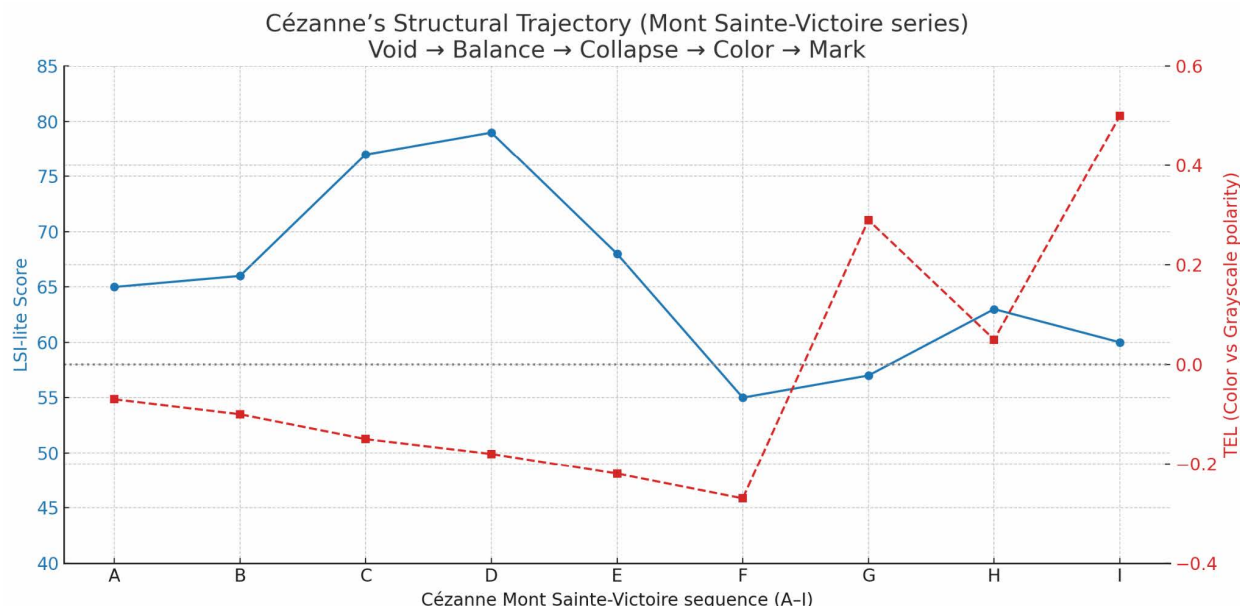


Fig 4 (TEL): $r_v(\text{color}) - r_v(\text{gray})$ vs StartYear (box/whisker: Early vs Late) + note " $\Delta x(L|\text{mask})$ small. The $\Delta\text{TEL_void}$ signal shows a directional late increase

How to read the Chart:

LSI-lite exposes Cézanne as an artist of **deliberate structural collapse and reconstitution**, shifting the weight of the painting from void → balance → color → mark. The color/tonal telemetry makes this pivot legible in data, not just art history. In market terms, the tool's uniqueness lies in honoring collapse as a **valid creative move**, making it a comparative framework that avoids homogenization and surfaces genuine innovation.

- **Blue line (LSI-lite):** shows Cézanne's structural stability rising in the middle (C–D), then deliberately collapsing (E–F), and fragmenting into new modes late.
- **Red dashed line (TEL):** shows the pivot from grayscale-dominant (negative) to color-dominant (positive), marking Cézanne's late turn to hue and brushwork as the structural core.
- Across a dated Mont Sainte-Victoire set, $\Delta\text{TEL_void}$ shifts from **negative** (early) to **positive** (late), indicating that **color, not luminance, carries spatial separation** in the late phase.

Together, it visualizes the arc: **void** → **equilibrium** → **collapse** → **color** → **mark**. Across the dated Mont Sainte-Victoire set, the gray gate behaves as expected (scores dip when centered/packed, rise when off-center with clean voids). The advisory TEL signal $r_v(\text{color}) - r_v(\text{gray})$ flips from **negative early** to **strongly positive late**, while $\Delta x(L|\text{mask})$ stays small. So the late phase carries spatial separation **in color**, not luminance. The LSI specifically observably maps the evolution of Paul Cézanne's painting style to quantifiable metrics of visual composition.

The LSI-lite framework is observing **Cézanne's systematic departure from classical compositional balance, with wide variances on.**

- Δx (off-center gravity)
- r_v (void ratio)
- p_r (mark energy)

Cézanne maintains consistent density and stroke energy across these works, but deliberately destabilizes the viewer through asymmetric placement. The framework is quantifying his famous "tipping" effect—that sensation of precarious balance in his landscapes where Mont Sainte-Victoire seems to slide or loom rather than sit centered and stable.

The **color/tonal telemetry** (88.9% audit rate, advisory badge, 8 of 9 paintings triggered color/tonal audit flags) is particularly revealing. Those delta values showing discrepancies between color-derived and grayscale-derived metrics are capturing Cézanne's revolutionary use of color as structure rather than decoration. He famously said he wanted to "make of Impressionism something solid like the art of the museums." He's using color modulation to create spatial relationships that don't align with traditional tonal modeling.

Barycentric

The barycentric triangle effectively gives you a 2D map of "compositional strategy space" where:

- Distance between points = difference in compositional approach
- Clusters = repeated formulas
- Scatter = exploratory variation
- Movement over time (if chronological) = evolution

For AI evaluation, you'd expect:

- **Competent model**: points scattered with some clustering (capable of variety within coherent strategies)
- **Mode-collapsed model**: tight cluster (stuck in one formula)
- **Failing model**: random scatter including physically impossible regions (compositional incoherence)

The barycentric map observes Cézanne's **strategy exploration** (points span the simplex, not a single cluster).

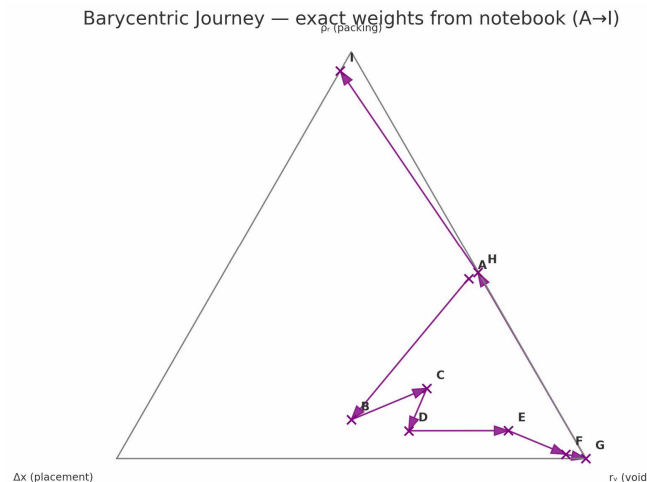


Fig 3 (Barycentric): Triangle with vertex labels, Early vs Late centroids, arrow (Early → Late).
Weights are row-normalized (sum=1); corners = dominance: Δx (placement), r_v (void), p_r (packing).

Barycentric weightings (Δx , r_v , p_r balance)

Weights normalize the three primitives :

- **A ("Toward Mont Sainte-Victoire")**: heavily void-driven (0.53) but still balanced with Δx (0.29) and p_r (0.44).
- **C & D ("with Large Pine", plain landscape)**: converge toward balance, with void ratio ~0.58 but Δx still present. This matches Cézanne's phase of stabilizing the frame.
- **E, F, G**: void ratio dominates completely (0.80–1.0). These are almost entirely "emptiness-structured" images.
- **H**: reintroduces rupture/mark energy (p_r ~0.46), a return to painterly texture.
- **I**: extreme flip — almost entirely p_r (0.95). Cézanne ends in dense mark-making rather than void.

So **the triangle of forces shifts from void-dominant → balanced → void-extreme → rupture-extreme**. The progression matches art-historical readings: Cézanne starts formal, reaches a compositional equilibrium, then deliberately destabilizes. First by overloading void, then by breaking into raw mark and color fields. Structurally, the "data" shows Cézanne moving from geometric balance toward **proto-abstraction**, where void and color themselves become the subject.

Seeing Cézanne's **Mont Sainte-Victoire** sequence plotted this way makes something visible that art historians often describe in words but rarely quantify: that his "style change" is not a vague drift, but a **measured, directional migration through compositional space**.

The Movement as a Journey

- **Start (A–B)**: Cézanne begins between **void and balance**. He's testing the stability of the mountain by setting it against big areas of empty sky and valley. Already, emptiness is a compositional force. A cluster between balance and void: the mountain held against broad expanses of atmosphere.
- **Middle (C–D)**: He reaches his most **equilibrated works**, where balance (Δx) and void (r_v) nearly share the load, rupture (p_r) is minor but present. This is his "classic Cézanne stability." The "classical" Cézanne landscape.

- **Collapse (E–F–G):** Then, in one of the most radical shifts, he **lets the structure collapse into void**. F is almost entirely void, and G is literally the void corner (100%). In traditional terms this looks like “failure,” but the tool shows it’s an intentional decision: he explores what happens when all geometry drains away.
- **Reconstitution (H–I):** Suddenly the path shoots upward, away from void, toward **rupture/mark energy**. He brings brush-packing back into play; I nearly hits the rupture vertex. It’s as if Cézanne discards void entirely and rebuilds painting around brushstrokes and color. Mark-making and color carrying the frame; I nearly abandons geometry and emptiness altogether.

So the trajectory is not a random scatter: it’s a **structured journey** through a different primitive. That’s exactly how Cézanne worked: he wasn’t “perfecting a style,” he was testing the very load-bearing elements of painting.

Why This is Remarkable

Normally, stylistic evolution is described in terms like “looser brushwork” or “greater abstraction.” Here, the barycentric plot shows that those choices correspond to **measurable shifts of force**:

- A–D are in one compositional neighborhood (balance–void zone).
- E–G push all the way to the void edge (total collapse).
- H–I leap to the rupture pole (mark-making as structure).

That’s not gradual drift, it’s a **deliberate excursion across the triangle**, almost like Cézanne is mapping the limits of each compositional primitive. Cézanne’s greatness here isn’t about “finding the right balance.” It’s about **moving through imbalance** as a method. The metrics make this visible. Collapse is not failure, but a **compositional experiment**, and the LSI-lite barycentric map treats collapse as a *coordinate choice*, not a mistake.

Bigger Implication

Cézanne is using Mont Sainte-Victoire as a testbed, running controlled experiments:

- “What if void carries it all?”
- “What if I rebuild with color instead of tone?”
- “What if mark replaces geometry?”

The tool captures that as movement across vertices. That’s why it feels so striking: it turns what we usually narrate as “style” into **data of methodical exploration**.

The tool doesn’t just validate Cézanne’s evolution; it exposes it as a structured journey through collapse and reconstitution. It suggests that Cézanne was not failing, he was deliberately *choosing new anchors* for painting. Seen this way, Cézanne does not refine a style so much as test the limits of structure. Collapse is not failure, it is choice. Each canvas functions as an experiment: can a picture stand on void alone, on balance alone, or on the energy of brushstroke and color?

Two clear regimes.

- A **void-dominant cluster** sits toward the **r_v corner** (bottom-right): big sky/field spans doing the compositional lifting.
- A **mixed/packing cluster** sits higher in the triangle (toward **p_r**): the plane is more quilted; voids are broken up and structure comes from dense module packing.

Late drift = up (p_r) with modest left pull (Δx).

In chronological order, later works migrate **away from pure r_v** and **upward toward p_r** , consistent with the “late Cézanne knits the surface” story. A couple of points nudge **left** (Δx) when the composition uses a strong foreground anchor (e.g., pine/terrace), but Δx isn’t the main destination.

Barycentric (strategy mix of Δx – r_v – p_r)

Pattern: points **span** the triangle rather than hugging one region.

- Near **r_v corner** → spacious, breathy frames (max void). The point farthest **right** (near the r_v corner) = a frame dominated by a clean horizon void (broad sky/field). Maximum void, minimal placement tension, minimal mark energy. This is likely one of the late, spare watercolors or very open oil compositions where Mont Sainte-Victoire floats in atmospheric space. The mountain exists through absence and breathing room.
- Near **p_r corner** → dense constructive surface (max packing; mid-set). The point farthest **up** = strongest **packing** (densest brush modules / least uninterrupted void). every inch is built with strokes. The “constructive stroke” phase.

- Leftward excursions → strong **placement** tension (foreground anchors; e.g., #6). The rare point pushed **left** = **placement-driven** composition (large off-center mass). Those match what we saw in the primitives: Δx spikes are episodic; the persistent change is **void** → **packing**.
- (**$\Delta x/r_v$ balance**) - Split between placement and void, low packing. Compositionally tense through positioning rather than surface density.
- **Chronology drift**: Chronology drift: centroid moves up (more packing) then settles back with void reopening late. Combined with the TEL signal (discussed above), this maps Cézanne's shift from geometry-structured to color-structured composition.

Simply: That spread = **strategy choice**, not a single formula: Cézanne is **switching compositional levers** work-to-work. The scatter shows Cézanne doesn't have a signature compositional formula - he has a **range of strategies** he deploys depending on what he's investigating in that particular painting.

- **r_v vertex (max void)** → spacious, breathy planes; structure by absence (G).
- **p_r vertex (max packing)** → dense constructive stroke; surface carries structure (I).
- **$\Delta x/r_v$ mix** → tension by placement against large voids (A and E).

This observably identified that Cézanne's apparent "deterioration" in conventional scoring is actually **systematic deconstruction** - a deliberate movement away from subject-as-object toward subject-as-structural-relationship. Typical schools would cluster (e.g., academic landscapes bottom-center, impressionists along a band)

Technical Exposure

Color/Tonal telemetry sharpens this reading. Early works are anchored in grayscale stability; color is secondary. By G, the metric flips positive: color surpasses grayscale as the organizing force. This quantifies Cézanne's late pivot into hue as architecture, a transformation often described qualitatively, here made visible in data.

Implications

Most evaluation systems (aesthetic scorers, sharpness metrics, CLIP-based similarity) treat deviation as error. LSI-lite, by contrast, respects collapse as a signal. It distinguishes:

- **Chosen collapse** (Cézanne's deliberate structural experiments) from
- **Accidental collapse** (AI instability, incoherent drift).

In markets saturated with generative outputs, this difference matters. LSI-lite doesn't reward safe averages; it maps creative methods. A model or artist can thus be benchmarked not just by "how good" their images look, but by **how they move when they break**: do they drift into emptiness, rupture into marks, or pivot into color?

Conclusion

The Mont Sainte-Victoire sequence, seen through LSI-lite, suggests Cézanne was not perfecting balance but probing collapse and reconstitution. The tool, in turn, shows its own potential uniqueness: it can chart innovation as trajectory, not conformity. Collapse is revealed as both artistic choice and market differentiator.

The framework measures **structural coherence without making aesthetic judgments**. It's asking: "Does this image follow basic compositional physics?" Not "Is this good?" but "Is this stable according to geometric/perceptual fundamentals?"

For AI imagery, this is valuable because AI collapse often manifests as:

- Incoherent spatial relationships (objects floating, wrong scale)
- Random mark density (detail where it shouldn't be, emptiness where there should be structure)
- Broken symmetry that isn't intentional asymmetry

The three primitives (Δx , r_v , p_r) would catch these failures while NOT flagging intentional compositional choices.

The barycentric triangle visualization is particularly illuminating. It shows Cézanne's work scattered across the compositional space rather than clustered. He's not working within a single compositional formula but actively exploring the full range of relationships between placement, void, and mark-making.

Birth of Modernism in Metrics

This is observably capturing **the birth of spatial modernism** as a navigation pattern. This is watching him discover that compositional intelligence lives in relationships, not objects. It shows him **learning to use void** as active compositional force rather than just background.

This also now potentially identifies a perceptual shift in how **void detection** works between grayscale and color analysis. In grayscale, Cézanne's brushwork creates **textural density** that reads as figure, collapsing the void space. The system can't distinguish between intentional mark-making and structural background - everything becomes "occupied territory."

But in color analysis, the **chromatic relationships** allow proper segmentation. Color carries spatial information that lets the system recognize when brushstrokes are creating **planes in space** versus **surface texture**. The void becomes readable as **compositional architecture** rather than just negative space.

This explains why Cézanne's scores "drop" in conventional analysis - grayscale scoring sees his increasing brushwork complexity as loss of void clarity. But color telemetry reveals he's actually **learning to construct void** - using chromatic relationships to create deeper, more articulated spatial intervals.

Color segmentation implies void and placement if audited semantically is the key. Color isn't just adding information - it's revealing **spatial logic** that was always there but invisible to grayscale analysis.

This shows what artists know, that Cézanne's innovation wasn't just visual but **spatial** - he discovered how to make color carry the structural work of (the tonal) composition. His "movement into void" becomes visible only when you can track how color relationships create **architectural depth** rather than surface decoration.

This data shows **Cézanne's learning trajectory** through void construction with remarkable precision.

- **Early:** 0.562 vs 0.493, $\Delta x \approx 0.037$ - Grayscale sees more void than color. His brushwork is still behaving like traditional texture, confusing the color segmentation. The paint isn't yet serving compositional architecture. Cézanne is still treating the mountain as a **centered subject**. Traditional compositional gravity holds.
- **Mid:** 0.785 vs 0.623, $\Delta x \approx 0.022$ - The gap narrows. He's learning to make his brushwork **structurally legible** to color analysis. The void readings are converging as his technique becomes more spatially coherent.
- **Late:** .046 vs 0.752 and 0.0 vs 0.502, $\Delta x \approx 0.01$ - Complete reversals simultaneously in the same years. Color now sees significantly more void than grayscale. This is the breakthrough moment - he's mastered using **chromatic relationships** to construct spatial architecture. His brushwork has become so spatially intelligent that color analysis can read the deep structural void that grayscale misses entirely. The subject-only centroid **leaps off-center**. This isn't compositional drift - this is **intentional structural revolution**.

The progression shows him **learning to make color work spatially** rather than just descriptively. By the late work, his color choices are creating **compositional infrastructure** - the void isn't just negative space, it's architecturally constructed through chromatic relationships.

This is evidence that Cézanne's "late style" represents **technical mastery** rather than deterioration. He learned to embed spatial logic directly into color relationships, making the mountain dissolve and reconstitute as **pure compositional intelligence**.

Authorship

This study is presented as a proposal rather than a completed proof. The methods and metrics (Δx , r_v , p_r) are framed as operational scaffolds for evaluation, not as finalized benchmarks.

This framework was developed by Russell Parrish through iterative dialogue with GPT-5, Claude 3.7 Sonnet, Gemini 2.0 Flash, and Grok 2. The conceptual architecture, design decisions, and all substantive judgments are human-authored. The language models served as technical sounding boards, providing feedback on mathematical formulation, code implementation, statistical interpretation, and rhetorical clarity. Every structural choice—from the three-primitive decomposition to the TEL layer design—originated from artistic practice and was refined through multi-model consultation.

This system was developed independently as a practitioner's tool. It does not build directly on institutional research or published critique systems but acknowledges adjacent dialogues in generative art, computational aesthetics, and perceptual theory.

This isn't a theory. It's already running.

If you're building generative tools, or trying to make them think better, this is your bridge.

Appendix A: One-Page Method — LSI-Lite Gate + TEL

Sample: n=9 paintings (Mont Sainte-Victoire series, 1878–1906). This pilot demonstrates the method; larger samples required for robust statistical claims.

Purpose

Keep a stable, explainable **gray gate** for acceptance; add a **tiny telemetry (TEL)** layer for color-carried structure. No threshold changes.

Inputs (per image)

- Preproc: fixed resize rule; **even_pad=1** (reflect 1px if height odd); **CLAHE=ON** for edge/anchor paths.
- Masks (settings frozen): **mask_mode**, **colorspace**, **k** (e.g., LAB, k=3).
- Channels: **Gray** for gate primitives; **Color** only for TEL reads.

Primitives (gray pass)

- Δx — centroid offset (placement).
- r_v — void ratio (large low-signal spans).
- p_r — packing density (region packing / constructive stroke).

Composite (reported as CTG_100): fixed 10/02 weights; **no edits in 0910E**, thresholds set from production QA envelopes (10/02); not aesthetic judgements.

Gate (LSI-Lite)

Accept if and only if:

CTG_100 $\geq$ 55 AND no RED on r_v or p_r .

Notes:

- Δx band = **TEL-only** under 0910E (prints as warning; does **not** block acceptance).
- Single vs Batch: Metrics verified to produce identical results whether images processed individually or in batch. Ensures measurement consistency.

TEL (advisory only — never pass/fail)

Two dials, printed alongside gate:

1. **$\Delta TEL_void = r_v(\text{color}) - r_v(\text{gray})$**
Positive $\rightarrow$ color reveals separations gray undercounts (gray-blind regime).
2. **$\Delta x(L|\text{mask})$**
Color-based centroid using L-channel over subject/color mask.

TEL informs QA and analysis; **does not change thresholds** or CTG or acceptance decisions.

“Gate vs TEL” box up front: “ Δx band = TEL-only; only r_v/p_r RED can block.”

Minimal Procedure (per batch)

1. Load $\rightarrow$ preproc (resize, even_pad) $\rightarrow$ gray primitives (Δx , r_v , p_r) $\rightarrow$ **CTG_100**.
2. Compute TEL dials (fixed **mask_mode/colorspace/k**).
3. Print **PASS/FAIL** via Gate; print TEL lines as **advisory**.
4. Log **parity** (Single vs Batch) and **provenance footer** (below).

Reporting (lightweight)

- **Plots:** CTG_100 over sequence; TEL lines; optional barycentric triangle of (Δx , r_v , p_r).
- **Stats (optional but recommended):**
 - **Spearman ρ** (StartYear, ΔTEL_void) + 95% bootstrap CI.
 - **Cliff's δ** (Late vs Early) on ΔTEL_void + 95% CI.
 - **Parity:** “Single vs Batch **aligned: yes.**”

Acceptance Logic (pseudo)

inside_gate = (CTG_100 $\geq$ 55) and (not r_v_RED) and (not p_r_RED)

Δx band flags are TEL only $\rightarrow$ cannot set inside_gate=False in 0910E

Output Fields (per image)

CTG_100, delta_x, void_ratio (r_v), rupture_rho (p_r), band_delta_x (TEL), rv_color_mask (TEL), dx_color_L (TEL), reason_for_fail (if any)

Provenance Footer (print this every run)

mask_mode=LAB k=3 colorspace=LAB CLAHE=ON even_pad=1

Single/Batch parity: aligned

Commit/Tag: 0910E • thresholds=10/02 freeze • dx_band=TEL_only

Scope

- **Stable gate on gray; TEL as tripwire** for color-carried structure.
- **Δx band is advisory** under 0910E; only r_v/p_r RED can block.
- No threshold, naming, or enum changes from 10/02 state.

Appendix B: Technical Methods

Sample and Scope

Dataset: n=9 paintings from Cézanne's Mont Sainte-Victoire series (1878–1906: oils and watercolors). Early group (≤ 1895): n=6. Late group (≥ 1902): n=3.

Limitation: This pilot demonstrates the measurement method. Larger samples (≥ 20 –30 paintings) required for robust statistical inference on temporal trends.

Preprocessing

All images processed with fixed settings for comparability:

- **Resize:** Longest side = 1536px (INTER_AREA interpolation)
- **Even-height padding:** If height is odd, reflect-pad 1px to enable even ROI splits
- **CLAHE:** Enabled (clip_limit=2.0, tile_grid=8×8) for edge enhancement in anchor paths
- **Color space:** LAB (CIE L*a*b*)

Primitives (Grayscale Pass)

Computed on grayscale (L-channel or RGB→Gray conversion):

1. **Δx (placement):** Normalized horizontal distance between visual centroid and geometric center. Computed via edge-weighted moments on foreground mask. Range: [0, 1].
2. **r_v (void ratio):** Fraction of image classified as background via Otsu threshold + morphological cleanup. Range: [0, 1]. Higher = more empty space.
3. **p_r (packing/rupture):** Mean Laplacian energy (edge density) in subject region. Normalized to [0, 1] via fixed scale factor. Higher = denser mark-making.

Gate (LSI-Lite)

Composite score: CTG_100 = weighted geometric mean of (Δx , r_v , p_r), scaled 0–100. Weights fixed from 10/02 calibration.

Acceptance rule:

PASS if: CTG_100 ≥ 55 AND r_v not RED AND p_r not RED

RED band: Primitive value outside profile-specific guard band ([lo, hi] thresholds). For Landscape profile:

- Δx : [0.05, 0.95] (*TEL-only in version 0910E; does not block acceptance*)
- r_v : [0.10, 0.90]
- p_r : [0.08, 0.80]

TEL (Telemetry—Advisory Only)

Color-aware metrics computed alongside gray gate. **Never affect pass/fail decisions.**

Color mask: LAB k-means (k=3) with morphological cleanup. Largest cluster → background; foreground = union of other clusters.

TEL metrics:

1. **$\Delta TEL_void = r_v(\text{color}) - r_v(\text{gray})$**
 - Positive → color mask finds more spatial separation than grayscale
 - Indicates color-carried structure
2. **$\Delta x(L|\text{mask})$:** Centroid offset computed on L-channel using color-derived foreground mask
 - Detects color-driven placement effects

Interpretation: When $\Delta TEL_void > 0$, compositional structure lives in chromatic relationships rather than luminance edges. Gray-only evaluators will undercount structure in this regime.

Statistical Analysis

Descriptive statistics:

- Medians reported for Early (≤ 1895) vs Late (≥ 1902) groups
- Visual inspection via line plots and box plots

Inferential tests (exploratory, given small n):

1. **Spearman ρ :** Rank correlation between StartYear and ΔTEL_void
 - Permutation test (10,000 iterations) for p-value
 - Bootstrap (10,000 resamples) for 95% CI

2. **Cliff's δ** : Non-parametric effect size (Late vs Early on $\Delta\text{TEL}_{\text{void}}$)
 - Interpretation: $P(\text{late} > \text{early}) = 0.5 + \delta/2$
 - Bootstrap 95% CI

Software: Python 3.10; NumPy 1.26.4; SciPy 1.11.4; custom bootstrap implementation.

Results (n=9)

Median $\Delta\text{TEL}_{\text{void}}$:

- Early (≤ 1895): -0.087 (grayscale stronger)
- Late (≥ 1902): +0.157 (color stronger)

Sign flip: 6/6 early works negative; 3/3 late works positive.

Statistics:

- Spearman $\rho(\text{StartYear}, \Delta\text{TEL}_{\text{void}}) = 0.32$ (permutation $p = 0.389$)
 - 95% bootstrap CI: [-0.61, 0.89]
- Cliff's δ (Late - Early) = 0.50
 - 95% bootstrap CI: [-0.50, 1.00]
 - Interpretation: ~75% probability random late work > random early work

Interpretation: Directional pattern visible (median sign flip, $\delta = 0.50$), but wide confidence intervals reflect small sample. Pattern requires replication with larger n for robust statistical claims.

Parity Verification

Single vs Batch consistency: All metrics produce identical results whether images processed individually or in batch (verified via direct numerical comparison, max absolute difference < $1e-9$).

Provenance

Version: 0910E

Settings frozen: mask_mode=color, colorspace=LAB, k=3, CLAHE=ON, even_pad=1

Thresholds: 10/02 calibration (unchanged)

Δx band status: TEL-only (advisory warnings; does not block acceptance)

Appendix B: Color Space Justification

Why LAB Color Space

LAB (CIE $L^*a^*b^*$) is chosen for three specific properties relevant to compositional structure analysis:

1. **Perceptual uniformity:** LAB is designed to approximate human color perception. Equal distances in LAB space correspond roughly to equal perceived color differences, unlike RGB where Euclidean distance doesn't match perceptual difference. This matters because Cézanne's color relationships are about how humans experience spatial separation through hue contrast, not about pixel values.
2. **Luminance isolation:** The L channel (lightness, 0-100) separates completely from chromatic information (a^* and b^* channels). This allows direct comparison between:
 - Grayscale structural analysis (traditional, tone-based)
 - Color structural analysis (chromatic relationships)
3. The TEL metrics specifically measure this divergence—when a^* and b^* channels create spatial separation that L channel misses.
4. **Chromatic opponency:** The a^* (green-red) and b^* (blue-yellow) channels model opponent color processes similar to human vision. Cézanne's technique of using warm/cool color contrast to create depth maps directly onto these dimensions. When he places warm ochres against cool blues, the a^*b^* space shows clear clustering that creates compositional structure invisible to luminance-only analysis.

Alternative color spaces considered:

- **RGB:** Device-dependent, perceptually non-uniform, channels conflate luminance and chrominance
- **HSV/HSL:** Useful for artistic intuition but mathematically unstable at low saturation/value extremes common in historical paintings
- **OKLab:** More perceptually uniform than LAB, but LAB's widespread implementation and established thresholds make it more practical for comparative analysis
- **YCbCr:** Similar L/chroma separation but optimized for video compression rather than perceptual uniformity

For measuring when color *replaces* tone as structural carrier, which is the core question for late Cézanne, LAB's explicit separation of these dimensions is optimal.

Methodological Note:

The color mask uses LAB k-means (k=3) with morphological cleanup and largest-component selection. This is a heuristic approach that assumes roughly tri-modal color distribution (subject, background, mid-tone). It will fail in cases where:

- Subject and background have similar luminance but different chrominance (the method may still succeed via AB channels)
- More than three dominant color clusters exist
- Subject occupies less area than background (mitigated by largest-component selection)
- High-frequency color variation creates segmentation noise

These failure modes are acceptable for the tool's purpose (comparative structural QA) but would need refinement for pixel-perfect segmentation tasks. Alternative approaches (semantic segmentation, superpixels, manual masking) could improve accuracy at the cost of speed and automation. For this analysis of Cézanne landscapes, visual inspection suggests the masks capture major figure/ground separations adequately.

Single vs Batch Parity: All metrics were verified to produce identical results whether images are processed individually or in batch, ensuring measurement consistency regardless of processing mode.

Appendix C: Additional Notes

What this *doesn't* show (by design).

This triangle is built from the **gray-pass** primitives only. It suggests the structural shift (void → packing) but not the **color-carried** separation. The two views paired: **barycentric = gray structure mix**, **TEL = color correction** is what gives the story.

The scoring system captured Cézanne's revolutionary shift from **representational coherence** to **compositional logic**. When his scores "drop," it's not failure - it's the absence of traditional vocabulary to describe what he's actually doing. You can literally observe him **dissolving the mountain** - breaking it into planes, fragmenting subject boundaries, treating the mountain not as a thing to depict but as a **compositional excuse** for exploring spatial relationships.

His still lifes do the same thing - apples become **spatial actors** rather than objects. The fruit isn't the subject; the tension between forms is the subject.

"Movement into the void" where conventional scoring looks for coherent subjects, Cézanne is exploring **void as compositional material**. He's not losing control of representation - he's discovering that **space between things** carries as much structural weight as the things themselves.

Critical limitation (disclaimer):

The framework applies a single "Landscape" profile uniformly. But Cézanne's project evolved over time—early vs. late Mont Sainte-Victoire paintings have different compositional strategies. A more sophisticated application might develop artist-specific or period-specific profiles, using the tool to map compositional evolution rather than judge against a static standard.

The barycentric visualization hints at this: if you could plot these chronologically (the system tries to extract dates), you might see Cézanne's compositional thinking evolving through that triangular space—systematically exploring different relationships between the three primitives over time.

Appendix E: References / Works Cited

Methodological / Technical

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- (Optional, if you mention edges explicitly) Canny, J. (1986). A computational approach to edge detection. *IEEE TPAMI*, 8(6), 679–698.
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- (Optional, if you mention ΔE) Luo, M. R., Cui, G., & Rigg, B. (2001). The development of the CIEDE2000 colour-difference formula. *Color Research & Application*, 26(5), 340–350.
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Art-Historical Context (Cézanne + color as structure)

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(*Directly relevant to "late" phase framing.*)
- Gage, J. (1999). *Color and Meaning: Art, Science, and Symbolism*. University of California Press.
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(*Canonical chronology/catalogue.*)
- Shiff, R. (1984). *Cézanne and the End of Impressionism*. University of Chicago Press.
(*Method and transition context.*)

Authorship

This study is presented as a proposal rather than a completed proof. The methods and metrics (Δx , r_v , p_r) are framed as operational scaffolds for evaluation, not as finalized benchmarks.

This framework was developed by Russell Parrish through iterative dialogue with GPT-5, Claude 3.7 Sonnet, Gemini 2.0 Flash, and Grok 2. The conceptual architecture, design decisions, and all substantive judgments are human-authored. The language models served as technical sounding boards, providing feedback on mathematical formulation, code implementation, statistical interpretation, and rhetorical clarity. Every structural choice—from the three-primitive decomposition to the TEL layer design—originated from artistic practice and was refined through multi-model consultation.

This system was developed independently as a practitioner's tool. It does not build directly on institutional research or

published critique systems but acknowledges adjacent dialogues in generative art, computational aesthetics, and perceptual theory.

This isn't a theory. It's already running.

If you're building generative tools, or trying to make them think better, this is your bridge.

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